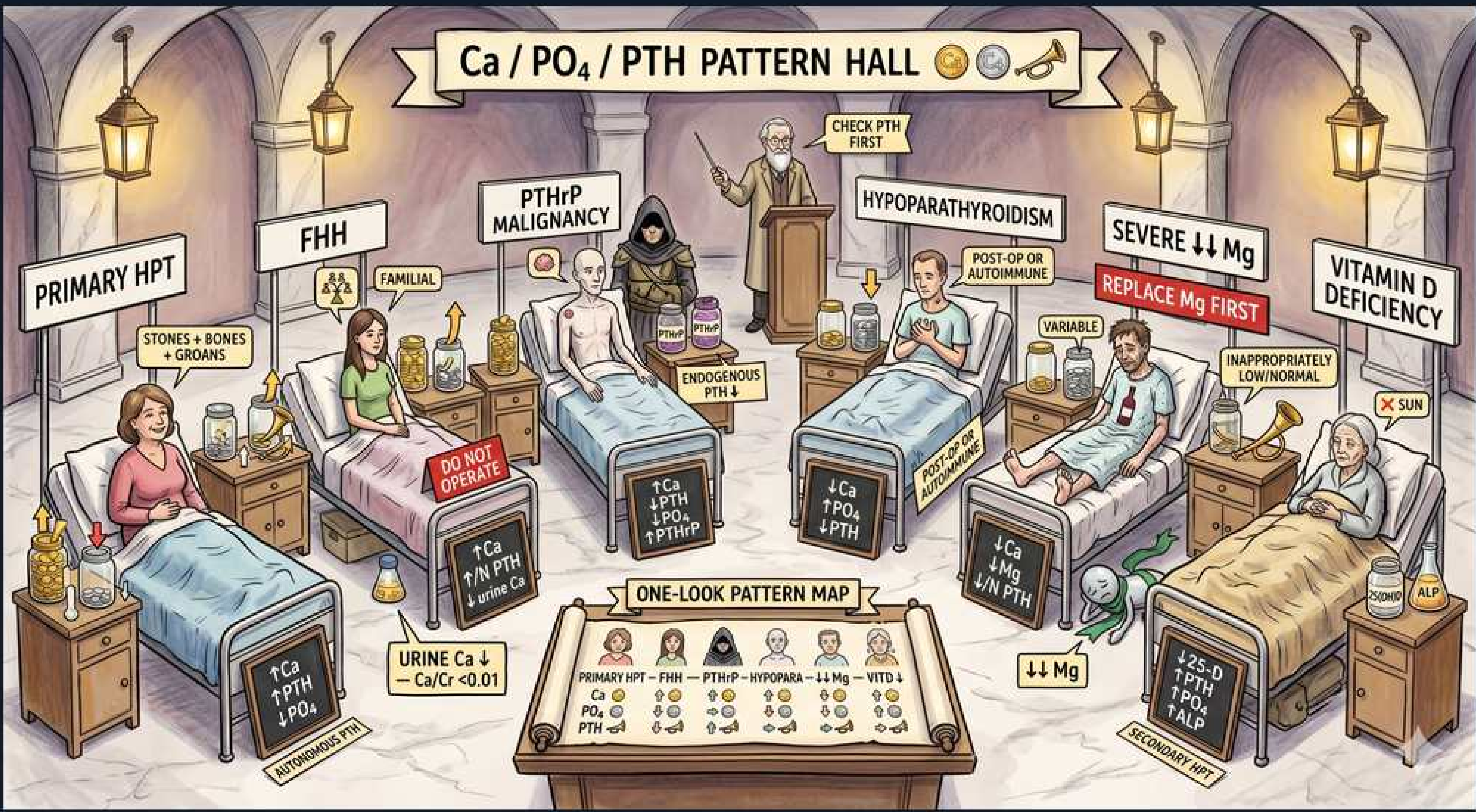
Calcium/PTH/VitD — The Ca:PO4:PTH Pattern Hall



1. Check PTH first (banner/lecturer)

WHAT YOU SEE

Center podium: 'CHECK PTH FIRST' with a lecturer.

WHAT IT ENCODES

For ANY abnormal calcium, PTH is the first and most discriminating test because it sorts disorders into PTH-dependent (PTH high/inappropriately normal) vs PTH-independent (PTH suppressed). In hypercalcemia: high/normal PTH = primary HPT or FHH; suppressed PTH = malignancy, vitamin D excess, granulomatous disease. In hypocalcemia: high PTH = appropriate response (vit D deficiency, CKD, PHP); low PTH = hypoparathyroidism or hypomagnesemia.

BOARD TRAP

Jumping to PTHrP, vitamin D levels, or parathyroid imaging (sestamibi) before checking PTH is the trap — those tests only make sense AFTER PTH has placed the patient in the dependent vs independent branch; PTH anchors the whole workup.

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2. Primary HPT bed (↑Ca ↑PTH ↓PO4)

WHAT YOU SEE

Leftmost bed: 'PRIMARY HPT — STONES + BONES + GROANS, ↑Ca ↑PTH ↓PO4'.

WHAT IT ENCODES

Primary hyperparathyroidism (usually a single parathyroid adenoma) = HIGH calcium, HIGH (or inappropriately normal) PTH, LOW phosphate (PTH is phosphaturic), and HIGH urine calcium. Most common cause of outpatient hypercalcemia. Symptoms: 'stones, bones, groans, psychiatric moans' — nephrolithiasis, osteitis fibrosa, abdominal pain/constipation, depression. Surgery (parathyroidectomy) is curative.

BOARD TRAP

The trap is operating on every hypercalcemia with high PTH — FHH gives the identical chemistry but has LOW urine calcium and is benign; always check the 24-hour urine Ca / Ca:Cr clearance ratio before parathyroidectomy.

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3. FHH bed (↑Ca ↑/N PTH, urine Ca low)

WHAT YOU SEE

Second bed: 'FHH — FAMILIAL, ↑Ca ↑/N PTH; URINE Ca — Ca/Cr <0.01; DO NOT OPERATE'.

WHAT IT ENCODES

Familial hypocalciuric hypercalcemia (FHH) = an INACTIVATING CaSR mutation (autosomal dominant): the receptor under-senses calcium, so PTH stays high/normal and the kidney avidly reabsorbs calcium → HIGH serum Ca but LOW urine Ca (calcium-to-creatinine clearance ratio <0.01). Lifelong, mild, asymptomatic, benign — no treatment needed.

BOARD TRAP

FHH mimics primary HPT chemically (high Ca, high/normal PTH) — the trap is sending the patient to parathyroidectomy, which fails to cure them. The LOW urine Ca/Cr clearance ratio (<0.01) is the discriminator that should stop the operation.

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4. PTHrP malignancy bed (↑Ca ↓PTH)

WHAT YOU SEE

Third bed: 'PTHrP MALIGNANCY — ↑Ca ↓PTH, ↓PO4; ENDOGENOUS PTH down'.

WHAT IT ENCODES

Humoral hypercalcemia of malignancy: a tumor (squamous cell lung, renal cell, breast, bladder) secretes PTH-related peptide (PTHrP), which acts on the PTH receptor to raise calcium and lower phosphate — but native PTH is SUPPRESSED (feedback from the high calcium). Most common cause of INPATIENT hypercalcemia. Other tumor mechanisms: osteolytic mets and 1,25-D production by lymphomas.

BOARD TRAP

Expecting PTH to be high in cancer-related hypercalcemia is the trap — endogenous PTH is LOW/suppressed; if PTH comes back HIGH, the diagnosis swings back to primary HPT, not malignancy. Order a PTHrP level only after PTH is found suppressed.

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5. Hypoparathyroidism bed (↓Ca ↑PO4 ↓PTH)

WHAT YOU SEE

Fourth bed: 'HYPOPARATHYROIDISM — POST-OP/AUTOIMMUNE, ↓Ca ↑PO4 ↓PTH'.

WHAT IT ENCODES

Hypoparathyroidism = LOW calcium, HIGH phosphate, and LOW (or inappropriately normal) PTH. Most common cause is post-surgical (neck/thyroid surgery); others are autoimmune (APS-1), DiGeorge (22q11), and infiltration. Symptoms: perioral numbness, tetany, Chvostek/Trousseau signs, prolonged QT. Treat with calcium + active vitamin D (calcitriol).

BOARD TRAP

Low Ca + high PO4 with LOW PTH = hypoparathyroidism; the trap is forgetting that the SAME low-Ca/high-PO4 picture with HIGH PTH means pseudohypoparathyroidism (PTH resistance) or CKD instead — the PTH direction flips the diagnosis.

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6. Severe ↓↓Mg bed (replace Mg first)

WHAT YOU SEE

Fifth bed: 'SEVERE ↓↓Mg — REPLACE Mg FIRST, ↓Ca ↓Mg ↓/N PTH, variable'.

WHAT IT ENCODES

Severe hypomagnesemia (Mg <1 mg/dL) causes low calcium with inappropriately LOW or normal PTH — a functional hypoparathyroidism, because Mg is required for PTH secretion AND PTH action. Causes: PPIs, loop/thiazide diuretics, alcoholism, diarrhea, aminoglycosides. The calcium will not correct until magnesium is replaced.

BOARD TRAP

Treating this hypocalcemia with calcium alone is the trap — it fails because the low magnesium blocks PTH; replace Mg first and the calcium and PTH normalize. Always check Mg in unexplained or refractory hypocalcemia.

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7. Vitamin D deficiency bed (↓Ca ↓PO4 ↑PTH)

WHAT YOU SEE

Rightmost bed: 'VITAMIN D DEFICIENCY — ↓25-D, ↓Ca ↓PO4 ↑PTH, ALP up, sun'.

WHAT IT ENCODES

Vitamin D deficiency = LOW calcium AND LOW phosphate, with HIGH (secondary, appropriate) PTH and ELEVATED alkaline phosphatase (from osteomalacia/rickets bone turnover). Low 25-OH vitamin D confirms it. Causes: poor sun/intake, malabsorption, liver/kidney disease. Both Ca and PO4 are low because secondary PTH wastes phosphate while gut absorption of both is impaired.

BOARD TRAP

The discriminator is phosphate: low Ca with LOW PO4 (and high PTH, high ALP) = vitamin D deficiency, whereas low Ca with HIGH PO4 points to hypoparathyroidism/PHP/CKD. Confusing these two high-PTH hypocalcemias on phosphate alone is the classic trap.

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8. One-look pattern map table

WHAT YOU SEE

Bottom table: 'ONE-LOOK PATTERN MAP' comparing Primary HPT, FHH, PTHrP, HypoPara, low Mg, Vit D across Ca/PO4/PTH.

WHAT IT ENCODES

Master comparison grid — each disorder has a signature Ca/PO4/PTH fingerprint: Primary HPT (↑Ca ↓PO4 ↑PTH), FHH (↑Ca N/↑PTH + low urine Ca), PTHrP malignancy (↑Ca ↓PO4 ↓PTH), Hypoparathyroidism (↓Ca ↑PO4 ↓PTH), Severe ↓Mg (↓Ca ↓/N PTH), Vitamin D deficiency (↓Ca ↓PO4 ↑PTH), Secondary HPT/CKD (↓-N Ca ↑PO4 ↑PTH). Pattern recognition pins the diagnosis at a glance.

BOARD TRAP

Memorizing the grid without the two discriminators is the trap: among HIGH-calcium states, urine Ca separates FHH (low) from primary HPT; among LOW-calcium states, phosphate separates vitamin D deficiency (low PO4) from hypoPTH/CKD (high PO4).

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9. Primary HPT column (↑Ca ↑PTH ↓PO4)

WHAT YOU SEE

Table column: Primary HPT row Ca↑ PO4↓ PTH↑.

WHAT IT ENCODES

Primary HPT signature: calcium HIGH, phosphate LOW, PTH HIGH. The low phosphate reflects PTH's phosphaturic action; this is the only hypercalcemic state where PTH is the driver and phosphate is low — pairs with high urine calcium and the stones/bones/groans presentation.

BOARD TRAP

Reading high Ca + high PTH as automatically primary HPT is the trap — FHH shares it; the phosphate is low in both, so use the URINE calcium (high in primary HPT, low in FHH) to separate them before surgery.

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10. Hypopara column (↓Ca ↑PO4 ↓PTH)

WHAT YOU SEE

Table column: hypoparathyroidism row Ca↓ PO4↑ PTH↓.

WHAT IT ENCODES

Hypoparathyroidism signature: calcium LOW, phosphate HIGH, PTH LOW. Without PTH there is no phosphaturia (PO4 rises) and no bone/renal calcium mobilization (Ca falls). This low-PTH profile is what separates true hypoparathyroidism from the high-PTH causes of low-Ca/high-PO4 (PHP and CKD).

BOARD TRAP

Assuming any low-Ca/high-PO4 patient is hypoparathyroid is the trap — only when PTH is LOW; if PTH is HIGH with the same Ca/PO4 it is pseudohypoparathyroidism (normal creatinine) or CKD (elevated creatinine).

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11. Vitamin D column (↓Ca ↓PO4 ↑PTH)

WHAT YOU SEE

Table column: vitamin D row Ca↓ PO4↓ PTH↑.

WHAT IT ENCODES

Vitamin D deficiency signature: calcium LOW, phosphate LOW, PTH HIGH (secondary), ALP HIGH. The LOW phosphate is the key contrast with hypoparathyroidism's HIGH phosphate — even though both can have low calcium, the phosphate direction and the PTH direction both differ.

BOARD TRAP

The trap is lumping all low-calcium states together — the LOW phosphate plus HIGH PTH uniquely flags vitamin D deficiency, distinguishing it from hypoPTH (high PO4, low PTH) and from CKD (high PO4, high PTH).

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12. Secondary HPT (CKD) reference

WHAT YOU SEE

Footnote appended to the 'ONE-LOOK PATTERN MAP' grid for SECONDARY HPT (CKD) — the extra row whose fingerprint (↓/N Ca, ↑PO4, ↑PTH) sits right next to vitamin D deficiency so the HIGH phosphate cell visibly separates the two high-PTH low-Ca states.

WHAT IT ENCODES

Secondary HPT of CKD: LOW/normal calcium, HIGH phosphate, HIGH PTH, LOW calcitriol, with elevated creatinine and FGF23. It is the appropriate parathyroid response to the calcitriol deficiency and phosphate retention of renal failure — distinct from vitamin D deficiency, which also has high PTH and low Ca but LOW phosphate.

BOARD TRAP

Distinguishing CKD secondary HPT from vitamin D deficiency is the trap — both show high PTH and low-ish calcium, but PHOSPHATE separates them (HIGH in CKD because the failing kidney retains it, LOW in vitamin D deficiency); the elevated creatinine confirms CKD.
